

Spam Email Classification Report

Objective

The objective of this project is to build a machine learning model capable of classifying messages as spam or non-spam.

Dataset Description

The dataset contains text messages and their corresponding labels.

Labels:

- ham = legitimate message
- spam = unwanted promotional or fraudulent message

Data Preprocessing

Text Feature Extraction

Machine learning algorithms cannot directly process text data.

Therefore, TF-IDF (Term Frequency – Inverse Document Frequency) was used to convert text into numerical feature vectors.

TF-IDF assigns higher importance to words that are significant within a message while reducing the importance of frequently occurring common words.

Dataset Split

The dataset was divided into:

- 80% Training Data
- 20% Testing Data

using `train_test_split()` with `random_state = 42`.

Model Selection

A Multinomial Naive Bayes classifier was selected because it is efficient and highly effective for text classification tasks such as spam detection.

Evaluation Metrics

The model was evaluated using:

- Accuracy

- Precision
- Recall
- F1-score

Accuracy measures the overall correctness of predictions.

Precision measures how many predicted spam messages were actually spam.

Recall measures how many actual spam messages were successfully identified.

F1-score balances precision and recall.

Results

Accuracy: 0.9668161434977578

Precision: 0.96(ham), 1.00(spam)

Recall: 1.00(ham), 0.75(spam)

F1-score: 0.98(spam), 0.86(spam)

Conclusion

The Naive Bayes classifier successfully classified text messages into spam and non-spam categories using TF-IDF features. The project demonstrates a complete NLP workflow including text preprocessing, feature extraction, model training, and evaluation.